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PolarDB for MySQL

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Sharded cluster compatibility (Limitless multi-master architecture)

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Product

As write pressure and data volume grow, MongoDB typically uses sharded clusters to scale writes horizontally, but this introduces significant operational complexity. PolarDB for MySQL Multi-master Cluster (Limitless) is built on shared storage and supports multi-write, multi-read, delivering horizontal write scaling without data relocation for document database (MongoDB-compatible) scenarios. This topic introduces the Limitless multi-master architecture, compares it with the MongoDB sharded architecture, and describes applicable scenarios.

Note The document database

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compatibility feature of Multi-master Cluster (Limits) is currently in canary release. If you have related requirements, please submit a ticket to contact us.

Background

The default deployment of the document database (MongoDB-compatible) feature uses a **one-primary-multi-read** architecture, and its usage experience is similar to a MongoDB replica set (Replica Set). However, in terms of implementation, PolarDB for MySQL has natural advantages:

- Read/write (RW) / read-only (RO) nodes share the underlying data, so read-only nodes have

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extremely low latency.

- The storage layer supports pay-as-you-go elastic scaling. For a more detailed comparison, see [Comparison with self-managed MongoDB](#) and [What is document database \(MongoDB-compatible\)](#).

As write pressure and data volume grow, the replica set architecture faces clear bottlenecks: only a single primary handles the write load, every member stores the full dataset, and scaling is limited to adding read capacity rather than write capacity.

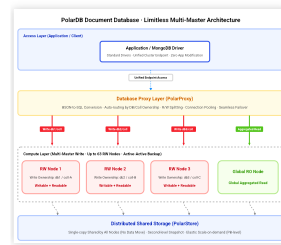
MongoDB typically uses sharded clusters to scale writes horizontally: data is partitioned into ranges by a shard key and distributed across different shards.

However, while the sharded architecture delivers write scalability, it also introduces significant operational complexity. Adding or removing shards relies on the Balancer to migrate and rebalance chunks, migration time depends on data volume, distribution, and throttling policies, and online workloads may be affected during migration.

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PolarDB Limitless multi-master architecture

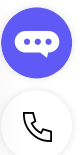


To eliminate the write bottleneck of the one-write-multi-read architecture, PolarDB for MySQL supports [Multi-master cluster \(Limitless\)](#). Multiple read/write (RW) nodes in a cluster share the data files in PolarStore, and the proxy layer routes requests to the corresponding RW node based on the access point of each database or data object.

Different databases or collections can be distributed across different read/write (RW) nodes for parallel writes, achieving cluster-level multi-write. A given database or collection is still written by a single read/write (RW) node at any moment. During scaling or load scheduling, the system mainly switches access points and routing relationships, without relocating data files between compute nodes.

In document database compatibility scenarios, applications continue to connect using standard

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MongoDB drivers and the unified cluster address. The proxy layer handles protocol translation and request routing, so applications do not need to be aware of the multi-master topology.

Comparison with the MongoDB sharded architecture

MongoDB achieves write scaling through sharded clusters, which consist of mongos routers, Config Servers, and multiple shards (each shard is typically a replica set with three replicas). Data is split into chunks by a shard key (Shard Key) and distributed across shards. The PolarDB for MySQL multi-master architecture supports multi-write, multi-read on shared storage, achieving the same write scaling goals with lower architectural and operational complexity:

Comparison dimension
Scale-out
Scale-in
Architecture components

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Resources and cost
Cross-node query
Data balancing
High availability

Core advantages and capabilities

Capability
Linear write scaling
Multi-master mutual backup
Elastic scheduling
Global aggregated read
Zero application changes
Serverless capability

Applicable scenarios

The following scenarios typically rely on MongoDB sharded clusters to solve write scaling problems. PolarDB Limitless multi-master architecture achieves comparable or even better scaling with lower architectural and

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operational complexity in the same scenarios.

- **SaaS multi-tenancy:**

organize data as one database per tenant. MongoDB sharding achieves tenant isolation through shard distribution, but tenant migration requires chunk relocation, which is time-consuming and causes jitter. With PolarDB Limitless, migration only switches ownership without moving data, achieving second-level migration with zero jitter and effectively avoiding issues with other tenants.

- **High-concurrency**

- writes for IoT and**

- logs:** split collections by business domain for concurrent writes. MongoDB sharding relies on Balancer rebalancing to handle hotspots, with a lagging response. With PolarDB Limitless, hot collections are scheduled to low-load read/write (RW) nodes in seconds, and write throughput grows near-linearly with the number of nodes.

- **Gaming global servers and zone-based servers:**

spread per-server databases during launch peaks and

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merge servers to reduce costs in the late stage. Merging servers with MongoDB sharding requires chunk relocation and node decommissioning, a long and high-risk process. With PolarDB Limitless, server merging only switches ownership without relocating data, imperceptible to online players, greatly reducing operational cost and risk.

- **Peak-and-trough businesses such as e-commerce promotions:**

MongoDB sharding requires predicting hotspots at the shard key design stage, and scaling for promotions depends on chunk migration warm-up, with elasticity measured in hours or even days. With PolarDB Limitless, read/write (RW) nodes are scaled out and hot collections distributed in minutes, then quickly consolidated after the promotion, compressing elasticity from days to minutes.

Summary

PolarDB Limitless multi-master architecture achieves horizontal write

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scaling without data relocation through shared storage, multiple read/write (RW) nodes, and a unified proxy layer, while maintaining MongoDB compatibility. Compared with traditional MongoDB sharded clusters, the architecture is simpler, operations are lighter, and elasticity is more agile.

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with self-

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managed

test report

MongoDB is this page

helpful?



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